Who Tips Their Taxi Driver?

DATA 400 Capstone Team Project Proposal
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1. Project overview

Tipping is an important part of taxi drivers' income, but tip amounts vary across trips. From a behavioral and labor economics perspective, we want to understand whether trip characteristics can help explain these differences. Using NYC yellow taxi data, we plan to explore tipping patterns and build machine learning models to predict tip percentages.
Our main research questions are:

How accurately can trip characteristics, such as fare, distance, time, location, and weather predict the tip percentage for card-paid NYC yellow taxi trips?
Which factors are most important for these predictions?

Previous studies have explored similar questions using machine learning. Huang (2024) used random forest classification and Lasso regression to predict NYC taxi tipping behavior, highlighting the challenge of trips with zero recorded tips and the importance of trip characteristics such as time, distance, and location. Similarly, Ding (2025) compared several machine learning models for predicting tip amounts in NYC yellow taxis and for-hire vehicles, finding that random forest performed best, with payment type and fare amount among the most important predictors.
Building on these studies, our project will incorporate weather conditions and evaluate tip percentage predictions over a 12-month period. Our analysis will focus on identifying patterns and evaluating predictive performance rather than establishing causal relationships. We hope our findings will help taxi drivers better understand variations in their earnings and provide useful insights for the NYC Taxi & Limousine Commission (TLC) and taxi payment technology providers.

2. Data

	TLC Trip Record Data	TLC Taxi Zone Lookup Table	Local Climatological Data (LCD)
Unit	One trip	Taxi zone	Hour
Key variables	Fare, tip, distance, pickup/drop-off time and zone, payment type, tolls, fees, vendor.	Zone, borough	Rain, temperature
Sample size	~1–2 million card-paid trips from 12 months	265 zones	~8,760 hours/station/year
Access	Public	Public	Public

Data Quality and Cleaning: We will focus on card-paid trips (`payment_type = 1`) and remove invalid records, including negative fares and unrealistic distances or durations. We will examine missing values, outliers, and reporting differences across months, documenting all cleaning decisions.
Target Variables: Our main target is tip percentage. Based on the exploratory analysis, we may also create a binary variable to classify trips using an appropriate tipping threshold.
Privacy and Limitations: Recorded tips may not capture cash tips, so our findings may not represent all passengers. We will use publicly available data and report findings at an aggregated level.

3. Analysis Plan

Goal: Predict tip percentages for NYC yellow taxi trips and identify which trip characteristics are most useful for prediction. We may also explore a classification task based on findings from EDA.

- **Data and Features:** Each observation represents one taxi trip. We will use fare, distance, duration, pickup and drop-off locations, time, tolls, fees, passenger count, and weather conditions (rain and temperature) as predictors. We will exclude `total_amount` because it includes the tip.
- **Exploratory Data Analysis (EDA):** Explore the distribution of tip percentages and how tipping varies by time, location, and trip type. We will also examine missing values, zero tips, and outliers to guide our cleaning decisions and identify a possible classification threshold.
- **Baselines:** Compare our models against two simple benchmarks: the overall average tip percentage and the average tip percentage by pickup location and hour, both calculated from training data.
- **Models:** Use linear regression, random forest, and gradient boosting to predict tip percentages. Depending on the EDA findings, we may also use logistic regression and random forest classification to predict whether a trip meets a selected tipping threshold.
- **Evaluation:** Split the data chronologically into training (first 9 months), validation (month 10), and testing (months 11-12). Evaluate regression models using MAE and RMSE, and classification models using AUC, precision, recall, and F1-score. Compare results with baselines, examine prediction errors across trip characteristics, and identify important predictors.
- **Robustness:** Check whether results remain consistent across seasons and different outlier-handling rules. If time permits, extend the analysis to green taxi trips.

4. Risks & Backup Plan

The main risk is that trip characteristics may not predict tipping behavior well because personal preferences are not captured in the data. If regression performs poorly, we will explore

classification using a threshold identified through EDA. We will also address the large dataset size through sampling or reducing the number of months analyzed if necessary.

5. Team Plan & Timeline

Checkpoint	Expected Deliverable	Lead	Support / Reviewer
Data Access (Oct 26 - 30)	Download data, create working sample, document sources & sampling	Leah	Linh, Gracie
Data & EDA (Nov 2 - 6)	Clean data, create data dictionary, analyze missing values, develop visualizations	Linh	Leah, Gracie
Baseline & Plan (Nov 9 – 13)	Build baseline models, define evaluation metrics & time-based split	Gracie	Leah, Linh
Initial Results (Nov 16 – 19)	Train models, evaluate performance, conduct error analysis	Gracie, Leah	Linh
Implications (Nov 23 - Dec 3)	Discuss stakeholder impacts, ethical concerns, & practical implications	Linh	Leah, Gracie
Complete Draft (Dec 7 – 10)	Finalize report, presentation materials, documentation, & reproducibility checks	All members	Peer review

Instructor Meetings: Tentatively November 5, November 12, & November 19.
Presentation: November 23, November 30, or December 3, depending on the assigned date.
Final Submission: December 10, 2026.
Our team will meet twice a week and communicate through a group chat. We will document major decisions and feedback in GitHub, and review each other's code and writing before submissions.

6. GitHub Repository

Link: https://github.com/gracienl/taxi-tips
Planned File Structure: `data/` (data processing and documentation), `notebooks/` (EDA and modeling), `figures/` (visualizations), `docs/` (report and project decisions), and `README.md` (project overview and reproduction instructions).

References

New York City Taxi and Limousine Commission. TLC trip record data. https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

Kantor, D., Casey, N. W., Menne, M. J., & Buddenberg, A. (2023). Local climatological data (LCD), version 2. NOAA National Centers for Environmental Information. https://www.ncei.noaa.gov/access/metadata/landing-page/bin/iso?id=gov.noaa.ncdc:C01689

Huang, H. (2024, January). Prediction of New York taxi tip behavior based on machine learning classification and regression methods. In *2023 2nd International Conference on Public Service, Economic Management and Sustainable Development (PESD 2023)* (pp. 686-698). Atlantis Press.

Ding, J. (2025, March). Predictive Modeling of Passenger Tipping Behavior for Yellow Taxis and For-Hire Vehicles in New York City Using Machine Learning Techniques. In *Proceedings of the 2025 5th International Conference on Applied Mathematics, Modelling and Intelligent Computing* (pp. 214-223).